PFMA-1: A 1-Hz, 150-kJ PULSED POWER SYSTEM FOR PLASMA FOCUS GENERATION *

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Abstract

PFMA-1 is a 150 kJ, 40 nH Mather-type plasma focus designed for repetition frequencies up to 1 Hz and dedicated to the neutron-free endogenous production of ^{18}F. PFMA-1 utilizes 32 capacitors, 11.1 µF each, charged in parallel at 30 kV. Snowplow calculations predict a 1.5 MA peak current and a 2.5 µs quarter-period for this device. To achieve low inductance and high peak current, one spark gap and four coaxial cables are employed per capacitor. Field-distortion, swing-cascade spark gaps were developed by R. E. Beverly III & Associates for this application (model SG-183). These switches mount directly onto the capacitor's modified Scyllac-type bushing and are enclosed by a low-inductance grounded shield. The rated peak current is 160 kA, the maximum charge transferred is ≈ 0.4 C/shot, the self-inductance is 26 nH, and the breakdown jitter is ≈ 1 ns at 30 kV and 60 kPa dry air. The switches are cooled by a combination of deionized water, SF6 and dry air. The estimated electrode lifetime is ≈ 0.1 Mshots. Special B-dot probes within each switch canister monitor the discharge of every capacitor. Simultaneous triggering of all capacitors is accomplished by a 32-channel PFL generator.

Some experimental results and waveforms obtained during preliminary testing of the facility are presented and compared with design values.

I. INTRODUCTION

In 2004, a joint research project by the Nuclear Engineering Laboratory of the University of Bologna and the Physics Department of the University of Ferrara was started, in the framework of the Research Agreement between the Alma Mater Foundation of the University of Bologna, the ULSS-12 of Venice and ABO-Project, in order to build and operate a 150 kJ, 1 Hz repetitive, Mather-type PF dedicated to the endogenous production of ^{18}F [1,2]. This device, PFMA-1, is now at an advanced state of realization [3] and some preliminary electrical [4] and plasma tests have already been accomplished.

II. PFMA-1

PFMA-1 is operated at 30 kV, with 32 GA32899 capacitors, 11.1 µF each, charged in parallel for a total bank energy of about 150 kJ. The overall inductance of the whole device in the short-circuit configuration is about 40 nH. This means that currents might flow in the device with a peak value of about 2.8 MA. Design of the device by optimization through 2D snowplow calculations has shown that the plasma load, at a pressure of about 10 Torr, should limit the maximum current to about 1.5 MA. The same calculations predict a quarter period of 2.5 µs and a slightly higher time-to-pinch. The number and model of capacitors have been chosen to take into account their contribution to the device inductance, mass and volume. A trade-off between high capacitance and low charging voltage at one extreme, and low capacitance and high voltage at the other, has been found, so that very fast damping of the current and voltage oscillations may be obtained in a discharge with the plasma load. This is of fundamental importance in increasing the lifetime of the capacitors; the rated voltage reversal at 36 kV for model GA32899 is about 60%; at 30 kV the over-voltage margin is much higher; in any case the first voltage waveforms, acquired with a fast capacitive probe, give a voltage reversal between 20% and 25% at most. The capacitor bank is charged in the constant current mode by an Alintel MF 12/30000 power supply, an SCR-controlled 400 kW power supply which can be regulated both in voltage (up to 30 kV) and current (up to 12 A). Parallel charging is achieved through a Distribution Box, designed and developed by R. E. Beverly III and Associates, which assures both channel-to-channel protection, in case of pre-firing or misfiring of some of the spark gaps, and voltage reversal snubbing towards the power supply. The components of this

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14. ABSTRACT **PFMA-1 is a 150 kJ, 40 nH Mather-type plasma focus designed for repetition frequencies up to 1 Hz and dedicated to the neutron-free endogenous production of 18F. PFMA-1 utilizes 32 capacitors, 11.1 ¦ÌF each, charged in parallel at 30 kV. Snowplow calculations predict a 1.5 MA peak current and a 2.5 ¦Ìs quarter-period for this device. To achieve low inductance and high peak current, one spark gap and four coaxial cables are employed per capacitor. Field-distortion, swing-cascade spark gaps were developed by R. E. Beverly III & Associates for this application (model SG-183). These switches mount directly onto the capacitor¡¯s modified Scyllac-type bushing and are enclosed by a low-inductance grounded shield. The rated peak current is 160 kA, the maximum charge transferred is ¡Ö 0.4 C/shot, the self-inductance is 26 nH, and the breakdown jitter is ¡Ö 1 ns at 30 kV and 60 kPa dry air. The switches are cooled by a combination of deionized water, SF6 and dry air. The estimated electrode lifetime is ¡Ö 0.1 Mshots. Special B-dot probes within each switch canister monitor the discharge of every capacitor. Simultaneous triggering of all capacitors is accomplished by a 32-channel PFL generator. Some experimental results and waveforms obtained during preliminary testing of the facility are presented and compared with design values.**		
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Distribution Box are completely oil–submerged and oil-cooled. A safety dump electromechanical device is used to ground the capacitors and dissipate charge. To achieve low inductance and high peak current, one spark gap and four coaxial cables are employed per capacitor. Field-distortion, swing-cascade spark gaps were developed by R. E. Beverly III and Associates specifically for this application (model SG-183). These switches mount directly onto the capacitor's modified Scyllac-type bushing and are enclosed by a low-inductance grounded shield. The rated peak current is 160 kA, the maximum charge transferred is ≈ 0.4 C/shot and the breakdown jitter is <1 ns at 30 kV and 60 kPa dry air. The breakdown time delay is <10 ns. Self-inductance was experimentally determined to be about 26 nH. The switches are cooled by a combination of deionized water, SF6 and dry air. The main electrodes are made of a sintered W-Cu alloy, while the triggering intermediate ring is made of AISI 304 stainless steel. The estimated electrode lifetime at rated conditions is ≈ 0.1 Mshots, however they can be easily replaced once worn out. Inside the grounded shield, a special B-dot probe is mounted and coupled to an Avago electrical to optical converter with BNC output connector. These B-dots provide both power for and signal to the electrical to optical converter which feeds 15-m long fiber optic cables. A 32-channel time jitter analyzer with optical to electrical converters compares the signals and produces an alarm in case a given switch has misfired for at least 3 consecutive shots, the allowed time gate for no alarm having been fixed at 15 ns maximum. The simultaneous triggering of all switches is accomplished by a 32-channel Pulse Forming Line generator (R.E. Beverly III and Associates model PG-107B). This unit employs a repetitive trigatron spark gap (SG-101M-75C, operated with nitrogen) which discharges 32 equal-length PFLs. A Glassman MK20P03.7 power supply charges these lines to 20 kV. The resulting triggering pulse has a FWHM of about 60 ns, an amplitude of about –40 kV and a fall time of about 7 ns. Four DS-2248 coaxial cables per spark gap, for a total of 128 parallel discharge cables, connect the capacitor bank to a circular cable collector of about 180 cm diameter. This collector consists of two AISI 304 2-cm thick stainless steel plates, insulated by a 1-cm thick Delrin slab and reinforced by 12 trapezoidal trusses each, to reduce possible membrane deformation due to electromagnetic forces during a shot. Multi-Contact sockets were used for connection of the coaxial cables. Since the gap between the ground part and the HV part of the collector is in air at atmospheric pressure, and the grounded shield of the spark gaps is instead filled with SF6 at 0.150 kPa relative pressure to suppress corona effects, the cables had to be impregnated with a special elastomer so to avoid SF6 leakage.
The main electrodes of the PF are made of OFHC copper. Their active length is 16 cm, the outer and inner diameters are 17 cm and 9.6 cm respectively, for an inductance contribution of about 18 nH.

The thickness of the active part of the ceramic insulator sleeve has been tested in the 4.6 – 6.25 mm range. A Rogowski coil has been placed inside the cable collector in order to measure the total current signal; a capacitive voltage divider, in the form of thin copper parallel foils,

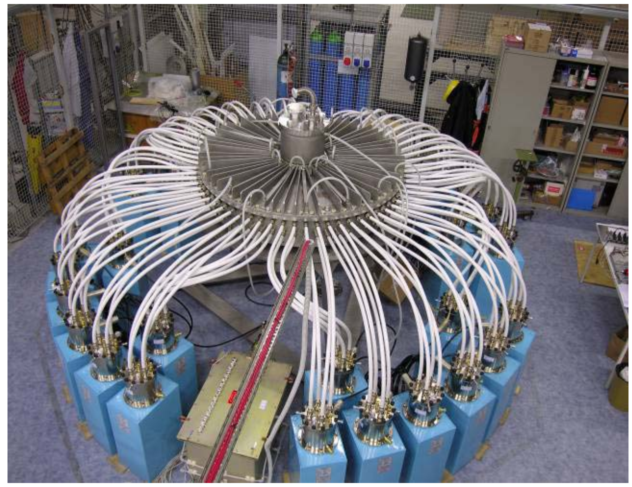

Figure 1. General view of the core of PFMA-1.

has been installed too, to measure the voltage averaged over the radial dimension of the collector. One single ground pole (resistance <4 Ohm) has been used to connect the ground part of the device, as well as all the other equipment, in a tree-structured topology thus avoiding any possible dangerous ground loops.

Figure 2. View of the interior of the vacuum chamber.

Heat management in PFMA-1 is a highly demanding task. Several fluids are used in combination to remove the thermal energy deposited in the various components of the device. Demineralized water, flowing in a partially open-circuit through an evaporative 150 kW cooling tower, will take care of the collector, electrodes and vacuum chamber. Closed-circuit DI water, refrigerated by two Termotek chillers, closed-circuit SF6 refrigerated by a special heat-exchange unit, and open-circuit dry air will contribute to the cooling of the spark gaps. The dielectric oil of the Distribution Box will be cooled by a Lytron heat-exchanger.

III. FIRST PLASMA-LOAD SHOTS

Initial short-circuit tests to evaluate the real, equivalent lumped parameters of PFMA-1 have been performed [4] and these confirm the design estimates. In the months of May and June 2007 PFMA-1 is under test with a real plasma load. More than 200 shots have been done and the associated electrical signals recorded. After an initial pump-down to about 20 µTorr, achieved with a BOC-Edwards maglev turbomolecular pump, the vacuum chamber has been statically filled with ^{4}He at pressures variable in the range 0.5 – 2 Torr. The gas is introduced in the vacuum chamber through a solenoid valve regulated by a MKS controller and Baratron pressure gauge. Pressure can be regulated to within ±0.05 Torr. For these preliminary tests, 31 capacitors have been used and charged at 18 kV, which is the minimum voltage for relatively small intra-channel jitter with the spark-gap switches operating at 0 kPa relative pressure. The discharged energy is about 55 kJ. Discharges were separated by 5 minutes waiting time, because all the cooling systems were disabled. This choice was dictated by our interest in determining the heat load imparted to the device. After a series of 10 shots, temperatures up to about 40 – 45 °C are reached in the massive AISI 304 stainless steel flange that closes the vacuum chamber and which is more or less directly exposed to the post-pinch plasma jets generated by the MHD instabilities. Visible surface damage by plasma contact has been detected over the inner surface of this flange. Studies and experiments with infrared cameras are presently under way to estimate the heat loads and the heat fluxes imparted to this flange.

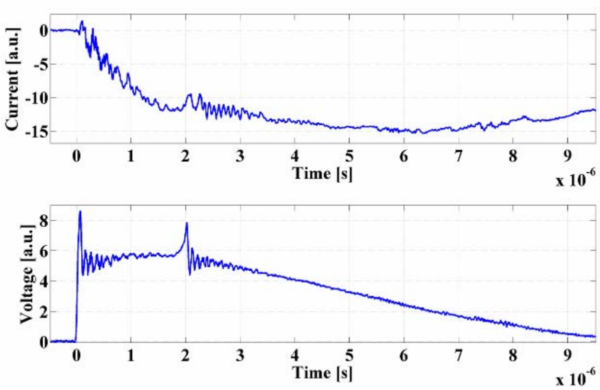

Figure 3. Current and voltage waveforms of a single pinch shot.

Preliminary results [5] indicate that at least about 15% of the bank energy is released in form of heat to the flange per shot. The distance between this flange and the pinch point is 14 cm. Figures 3 and 4 show the current and voltage waveforms for two shots at 1 Torr ^{4}He; in the former figure, one single pinch is clearly visible, in the latter figure, a pinch is immediately followed by another, the temporal distance between the two being about 250 – 300 ns. Figure 5 shows voltage and current waveforms for a 1 Torr ^{4}He shot which evidenced the structures of a triple (and maybe even quadruple) pinch. Multiple pinches are probably related to the operation at low pressures of PF devices [6].

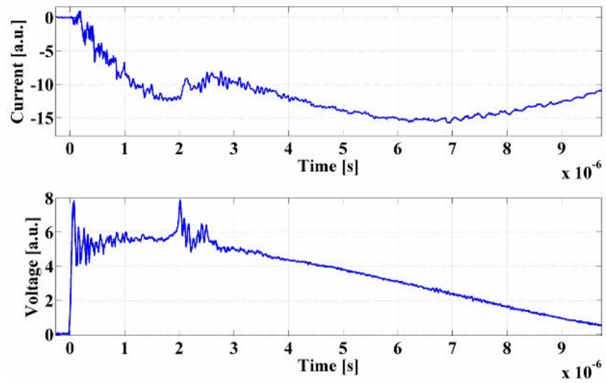

Figure 4. Current and voltage waveforms of a double pinch shot.

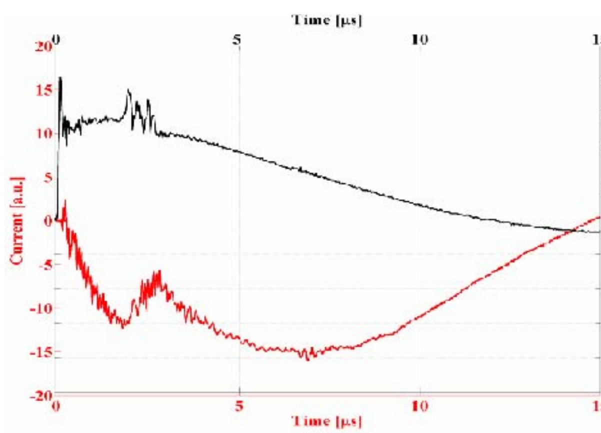

Figure 5. Current and voltage waveforms of triple pinch shot.

The duration of the whole multiple pinch phase for this case is slightly less than 1 µs. In all the cases of Figures 3 to 5, notice that the pinch occurs at 2 µs, much in advance with respect to the current maximum; this is due to the fact that in these tests the ^{4}He pressure was not optimized. All the voltage waveforms show that the typical over-voltage spikes at pinch-time are comparable in amplitude with those just before breakdown. Some noise in the current waveforms is detectable too; this is most probably due to the rather large jitter in the breakdown of the dry air in the spark-gap switches at 18 kV. An increase in the charging voltage would result in a much lower jitter. A test at 2 Torr and 25 kV as shown by the current trace in Figure 6, with an appropriate increase in the switch pressure, shows substantially reduced jitter sufficient to almost eliminate the superimposed noise. No pinch is visible, probably because some contamination of the filling gas, due to a previous opening of the vacuum chamber, had occurred. Figure 7 shows the voltage waveform corresponding to the current trace in Figure 6. The voltage reversal for PFMA-1 is clearly rather low and typically only 20 – 25%.

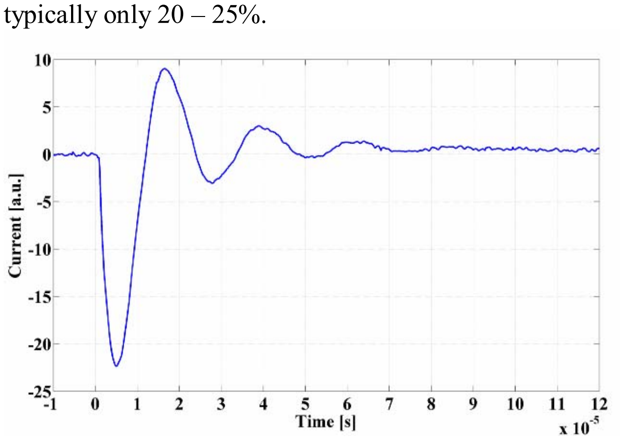

Figure 6. Current waveform at 25 kV.

This was a highly desirable goal during the design phase of this pulsed power system because, being much lower than the rated value, it increases the capacitor's lifetime. A fast damping of the oscillations is also evident from the waveforms, a result of the rather low inductance achieved by the design and development effort.

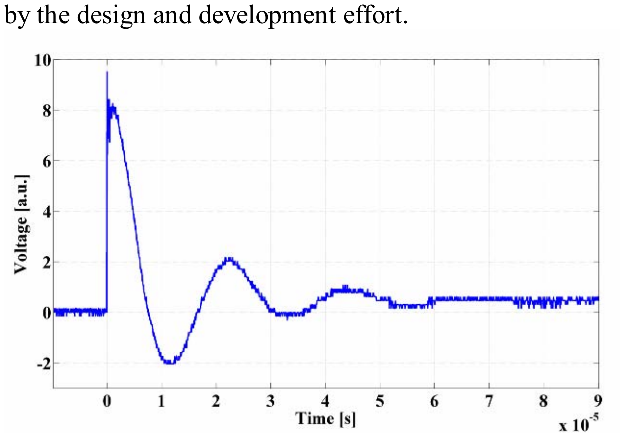

Figure 7. Voltage waveform at 25 kV.

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